

Three exposures. One world in color.

Reconstructing the Prokudin-Gorskii collection through channel alignment, exhaustive search and coarse-to-fine image pyramids.

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18 + 2

Provided + additional plates

6

High-resolution input scans

2.74 s

Slowest provided alignment



Handwritten NCC pyramid · original scan processed at 3741 × 9715

Recover the alignment, then recover the color.

The inputs are grayscale scans containing three vertically stacked exposures in blue, green and red order. The program divides each scan into three equal-height arrays, discards at most two trailing rows, keeps blue fixed and translates green and red. The final channel order is red, green, blue.

Stacked B / G / R → Split & normalize → Estimate G / R shifts → Compose RGB

Single-scale search

For each moving channel, try all 961 integer shifts in a ± 15 -pixel window. Compare a fixed interior region of blue against the shifted channel. Exclude a 12% scoring border on every side, expanding the exclusion when needed to keep every tested coordinate in bounds.

All candidates at a level use exactly the same reference pixels. This avoids favoring shifts that compare a smaller overlap or introduce wraparound.

Two baseline metrics

$$L_2 = \sqrt{\sum (B - C_{\text{shifted}})^2}$$

Choose the smallest distance. L_2 is sensitive to differences in brightness across color filters.

$$NCC = \frac{\sum (B \cdot C_{\text{shifted}})}{(\|B\| \|C_{\text{shifted}}\|)}$$

Choose the largest similarity. This is the normalized dot product in the handout, without mean subtraction. It compensates for multiplicative intensity changes, but not arbitrary color-dependent changes.

Convert unsigned integer pixels to floating point using their dtype range. Use Pillow only for reading, writing and resizing; NumPy performs the manual search and array operations. No automatic registration or pyramid-building library is used.

Start small. Make each step visible.

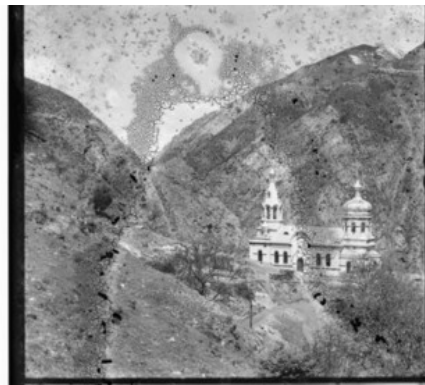
Every provided plate receives a low-resolution experiment using channels resized to 320 pixels wide. Both L2 and NCC run on the same pixels and search window. The three examples below show the separated exposures, unaligned composite and two aligned composites. Offsets here are in **low-resolution pixels**.

DETAILED EXPERIMENT 01

00056v.jpg



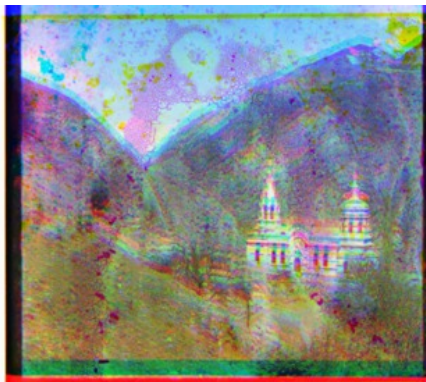
B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: G (1, 4) · R (0, 10); same for L2 and NCC.

The church and hillside become much easier to read after registration. The colored speckles remain: the separate source exposures contain different scratches and blotches. A translation cannot restore missing emulsion.

DETAILED EXPERIMENT 02

00125v.jpg



B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: $G(2, 4) \cdot R(1, 8)$; same for L2 and NCC.

The skyline, church and riverbank provide stable geometric structure. The two metrics select the same shifts at this reduced resolution. Narrow colored bands remain near the original plate edges.

00163v.jpg



B channel



G channel



R channel



Unaligned RGB



Single-scale L2



Single-scale NCC

Low-resolution shifts: G (1, -2) · R (1, -3); same for L2 and NCC.

Tree trunks and roof lines provide alignment cues. The negative vertical shifts are valid: these channels must move upward relative to blue. The two metrics again select the same low-resolution shifts.

All low-resolution experiments

Each shift cell lists G first, then R. Times are NCC / L2 in seconds. Plate numbers map to the labeled results below.

Plate	NCC shifts	L2 shifts	Time (s)	Images
Plate 01	(1, 4)	(1, 4)	0.383 / 0.237	NCC · L2
	(0, 10)	(0, 10)		
Plate 02	(2, 4)	(2, 4)	0.387 / 0.236	NCC · L2
	(1, 8)	(1, 8)		
Plate 03	(1, -2)	(1, -2)	0.378 / 0.256	NCC · L2
	(1, -3)	(1, -3)		
Plate 04	(0, 4)	(0, 4)	0.442 / 0.287	NCC · L2
	(3, 8)	(3, 8)		

Plate	NCC shifts	L2 shifts	Time (s)	Images
Plate 05	(-2, 5) (-4, 10)	(-2, 5) (-4, 10)	0.441 / 0.269	NCC · L2
Plate 06	(1, 5) (1, 11)	(1, 5) (1, 11)	0.382 / 0.259	NCC · L2
Plate 07	(2, 2) (3, 6)	(2, 2) (3, 6)	0.474 / 0.315	NCC · L2
Plate 08	(1, 5) (3, 9)	(1, 5) (2, 9)	0.416 / 0.281	NCC · L2
Plate 09	(2, 5) (2, 11)	(2, 5) (2, 11)	0.400 / 0.241	NCC · L2
Plate 10	(2, 5) (1, 10)	(1, 5) (1, 10)	0.409 / 0.244	NCC · L2
Plate 11	(1, 6) (1, 13)	(1, 6) (1, 13)	0.385 / 0.258	NCC · L2
Plate 12	(0, 7) (-1, 13)	(0, 7) (-1, 14)	0.391 / 0.249	NCC · L2
Plate 13	(1, 5) (1, 10)	(1, 5) (1, 10)	0.509 / 0.346	NCC · L2
Plate 14	(4, 7) (7, 14)	(4, 7) (6, 14)	0.451 / 0.267	NCC · L2
Plate 15	(1, 7) (1, 15)	(1, 7) (1, 15)	0.413 / 0.246	NCC · L2
Plate 16	(3, 6) (5, 13)	(3, 6) (5, 13)	0.404 / 0.256	NCC · L2
Plate 17	(2, 4) (2, 9)	(2, 4) (2, 9)	0.381 / 0.265	NCC · L2
Plate 18	(0, 7) (0, 11)	(0, 7) (0, 11)	0.389 / 0.244	NCC · L2

Large shifts. Small searches.

Explicitly build a list of progressively smaller images using bilinear downsampling by about two. Stop when the shorter channel dimension is at most 160 pixels. At the smallest level, search ± 15 pixels. At each finer level, multiply the estimate by the actual width/height ratios, round to integer pixels and refine within ± 2 pixels. Using actual ratios handles odd image sizes.

All 18 provided scans are processed at their original resolution. The same NCC metric and parameters apply to every image; no per-image tuning is used. A finer level tests only 25 candidates per moving channel, instead of exhaustively searching for shifts exceeding 100 pixels at full size.

A pyramid in motion

Intermediate estimates for 00458u.jpg; offsets are measured in pixels of the indicated level.

Channel size	G shift	R shift	Search radius
117 × 102	(0, 1)	(1, 3)	± 15
234 × 203	(0, 3)	(2, 6)	± 2
468 × 405	(1, 5)	(4, 11)	± 2
936 × 810	(1, 11)	(8, 22)	± 2
1871 × 1619	(3, 21)	(16, 44)	± 2
3741 × 3238	(6, 43)	(32, 87)	± 2

Output cropping removes only pixels without support in all three shifted channels. This is a geometric validity crop, **not** automatic detection of the original photographic borders; colored plate borders therefore remain visible.

Original-resolution offsets and runtime

Shifts are (x, y), positive right/down, applied to G and R relative to B. Timing covers both channel alignments, including pyramid construction, but excludes loading, compositing and image saving. Median: 0.089 s; range: 0.078–2.741 s on this Windows environment. No full-resolution brute-force timing comparison was performed.

Plate	G shift	R shift	Levels	Time (s)
Plate 01	(1, 6)	(1, 13)	3	0.078
Plate 02	(2, 5)	(1, 10)	3	0.081

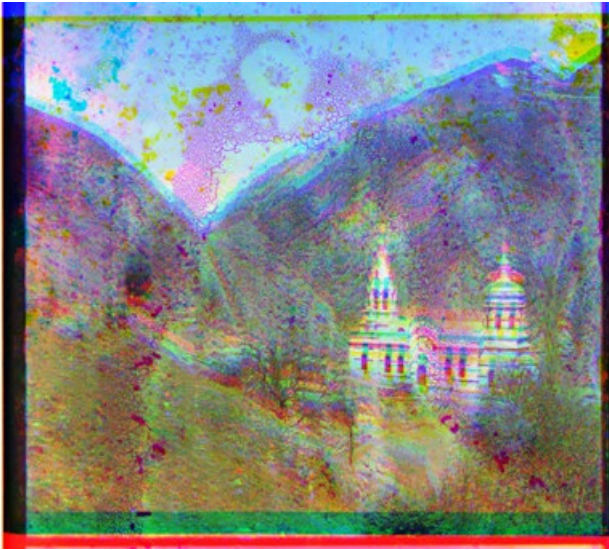
Plate	G shift	R shift	Levels	Time (s)
Plate 03	(1, -3)	(1, -4)	3	0.079
Plate 04	(6, 43)	(32, 87)	6	2.390
Plate 05	(-2, 6)	(-4, 13)	3	0.087
Plate 06	(14, 60)	(13, 129)	6	2.382
Plate 07	(20, 24)	(33, 71)	6	2.741
Plate 08	(2, 6)	(3, 11)	3	0.090
Plate 09	(2, 6)	(3, 13)	3	0.080
Plate 10	(2, 6)	(2, 13)	3	0.079
Plate 11	(1, 7)	(1, 16)	3	0.081
Plate 12	(0, 8)	(-1, 16)	3	0.096
Plate 13	(9, 55)	(12, 117)	6	2.473
Plate 14	(44, 77)	(77, 161)	6	2.417
Plate 15	(1, 8)	(1, 18)	3	0.091
Plate 16	(38, 71)	(63, 148)	6	2.459
Plate 17	(2, 6)	(3, 12)	3	0.086
Plate 18	(0, 8)	(0, 13)	3	0.086

04 / ALL PROVIDED IMAGES

The complete reconstruction set.

Each pair shows the unaligned composite and the NCC pyramid result. Click an aligned image to open the saved result at original pixel scale, after the common-support crop. Web previews are at most 900 pixels wide.

00056v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 6) · R (1, 13) · applied at original resolution

Source damage remains visible as colored blotches.

PLATE 02 / PROVIDED

00125v.jpg

400 × 1024 input

0.081 s alignment



Before · unaligned RGB



After · NCC pyramid

G (2, 5) · R (1, 10) · applied at original resolution

PLATE 03 / PROVIDED

00163v.jpg

387 × 1024 input

0.079 s alignment



Before · unaligned RGB



After · NCC pyramid

G (1, -3) · R (1, -4) · applied at original resolution

PLATE 04 / PROVIDED

00458u.jpg

3741 × 9715 input

2.390 s alignment



Before · unaligned RGB



After · NCC pyramid

G (6, 43) · R (32, 87) · applied at original resolution

PLATE 05 / PROVIDED

00804v.jpg

396 × 1024 input

0.087 s alignment



Before · unaligned RGB



After · NCC pyramid

G (-2, 6) · R (-4, 13) · applied at original resolution

PLATE 06 / PROVIDED

01007a.jpg

3800 × 9715 input

2.382 s alignment



Before · unaligned RGB

G (14, 60) · R (13, 129) · applied at original resolution



After · NCC pyramid

PLATE 07 / PROVIDED

01047u.jpg

3741 × 9656 input

2.741 s alignment



Before · unaligned RGB

G (20, 24) · R (33, 71) · applied at original resolution



After · NCC pyramid

01164v.jpg



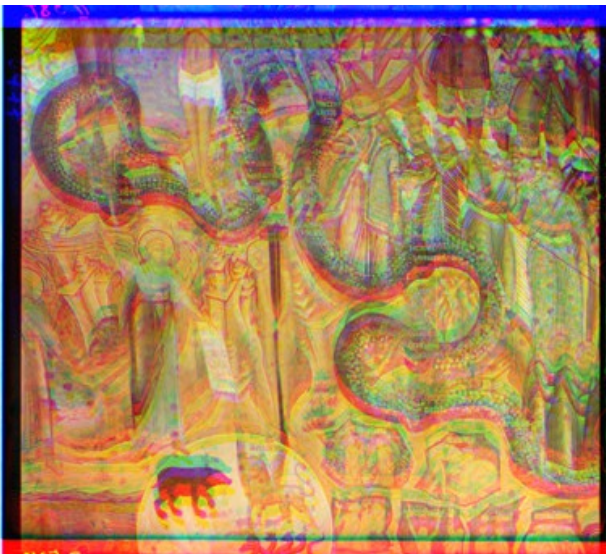
Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (3, 11) · applied at original resolution

01269v.jpg



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (3, 13) · applied at original resolution

01522v.jpg



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (2, 13) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01597v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 7) · R (1, 16) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01598v.jpg



Before · unaligned RGB



After · NCC pyramid

G (0, 8) · R (-1, 16) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

01657u.jpg



Before · unaligned RGB



After · NCC pyramid

G (9, 55) · R (12, 117) · applied at original resolution

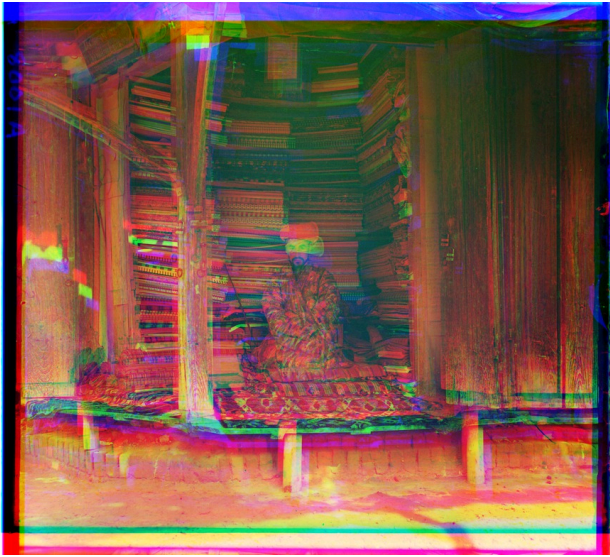
Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

PLATE 14 / PROVIDED

3731 × 9695 input

2.417 s alignment

01725u.jpg



Before · unaligned RGB



After · NCC pyramid

G (44, 77) · R (77, 161) · applied at original resolution

PLATE 15 / PROVIDED

397 × 1024 input

0.091 s alignment

01728v.jpg



Before · unaligned RGB



After · NCC pyramid

G (1, 8) · R (1, 18) · applied at original resolution

01861a.jpg



Before · unaligned RGB



After · NCC pyramid

G (38, 71) · R (63, 148) · applied at original resolution

Inspect faces, hands and foliage for local residual color fringes; a single translation cannot model independently moving subjects.

PLATE 17 / PROVIDED

10131v.jpg

400 × 1024 input

0.086 s alignment



Before · unaligned RGB



After · NCC pyramid

G (2, 6) · R (3, 12) · applied at original resolution

PLATE 18 / PROVIDED

31421v.jpg

398 × 1024 input

0.086 s alignment



Before · unaligned RGB



After · NCC pyramid

G (0, 8) · R (0, 13) · applied at original resolution

Beyond the supplied dataset.

Two additional three-frame grayscale JPEGs were downloaded from the Library of Congress, rather than using its finished color restorations. Both use exactly the same pipeline and parameters as the provided set. These examples are low-resolution scans; the large-image demonstration comes from the six supplied high-resolution scans.

ADDITIONAL COLLECTION EXAMPLE

00166v.jpg

V Malorossii (In Little Russia [i.e. Ukraine])



Before · unaligned RGB



After · NCC pyramid

G (2, 3) · R (4, 12) · 0.085 s

Official catalog record · Original grayscale scan · Single-scale NCC · Single-scale L2

ADDITIONAL COLLECTION EXAMPLE

00637v.jpg

Probnaiia shtolnia (Trial adit; iron ore piled in front)



Before · unaligned RGB



After · NCC pyramid

G (-2, 4) · R (-5, 10) · 0.094 s

Official catalog record · Original grayscale scan · Single-scale NCC · Single-scale L2

06 / BELLS & WHISTLES

Beyond alignment. Three automatic adjustments.

Alignment brings the exposures together, but does not remove the photographic frame, expand a faded tonal range, or correct a global color cast. I implemented automatic cropping, automatic contrasting and automatic white balance with NumPy. Pillow handles image input, output and preview resizing. These libraries expose the pixel operations directly and require no additional dependencies.

All 20 scans use the same parameters. The original scans are verified against their saved SHA-256 hashes and recomposed with the baseline shifts before enhancement; processing does not start from a compressed output JPEG. Each individual comparison below applies only its named method to the same aligned baseline. Combined results apply cropping, then white balance, then contrast, so frame pixels do not influence the final color and intensity statistics.

1. Automatic cropping

What it does: inspect rows and columns from each of the four edges. For every candidate boundary, measure the median signed RGB change across the central 60% of the line, then take the largest absolute channel change. This favors a coherent border-to-scene transition while cancelling inconsistent texture changes. The central span reduces interference from perpendicular borders.

A transition must exceed both 0.035 normalized intensity and four times the typical interior transition. At least 60% of the preceding strips must look border-like: their within-channel 90th–10th percentile spread is below 0.12, all channel medians are above 0.9 or below 0.1, or the range across their channel medians exceeds 0.35. The innermost qualifying transition defines that side's crop. The outer 15% is a **search limit, not a fixed margin**; a side with no qualifying transition is retained. Subsampling wide strips keeps profile estimation inexpensive.

Why use it: physical frames can be white, black or colored, so brightness alone is insufficient. Combining color uniformity with a spatial transition removes detected frame regions after the separate common-support crop used by alignment.



Before · aligned RGB



After · detected-border crop

Example 00163v: detected left / top / right / bottom margins = 9 / 10 / 13 / 8 pixels; 89.3% of the aligned image area retained.

Limitation: this is a heuristic for roughly rectangular frames. A flat sky or a strongly colored scene edge can resemble a frame; damaged, textured or very wide borders may remain. Retained area is a size measurement, not proof of correct segmentation.

2. Automatic contrasting

What it does: collect intensities across all RGB channels, estimate their 1st and 99th percentiles, and apply one shared linear mapping to every channel.

$$I' = \text{clip}((I - p_1) / (p_{99} - p_1), 0, 1)$$

Why use it: percentile endpoints are less sensitive than the absolute minimum and maximum to isolated scratches and extreme pixels. A common mapping avoids three independent channel stretches that could introduce an additional color cast. It preserves channel ordering, although subtracting a common offset can change saturation. Images with a negligible percentile range are left unchanged to avoid division by zero.



Before · aligned RGB



After · 1–99% contrast stretch

Example 00458u: shared endpoints 0.0039 and 0.9647; 0.90% of channel samples lie outside those endpoints.

Limitation: clipping sacrifices extreme shadow and highlight detail and can emphasize blemishes. The recorded clipping fraction counts channel samples, not whole pixels. This global adjustment does not recover missing detail or adapt separately to locally bright and dark regions.

3. Automatic white balance

What it does: use the gray-world assumption that a sufficiently varied scene has an approximately neutral average reflectance. Estimate the illuminant's relative RGB strength from the image's channel means μ . Set the neutral target t to their average and compensate with a diagonal channel scaling.

$$t = (\mu_R + \mu_G + \mu_B) / 3; \quad g_c = t / \mu_c; \quad I'_c = g_c I_c$$

Gains are limited to $[0.5, 2]$ to restrain amplification of weak channels; a channel with mean at most 10^{-7} keeps unit gain. If the corrected maximum exceeds one, divide the entire RGB image by that maximum, preserving the corrected channel ratios instead of clipping highlights independently.

Why use it: it explicitly separates illuminant estimation from color compensation, needs no manually selected neutral patch, and is easy to reproduce. A weak average channel receives more gain; a dominant one receives less.



Before · aligned RGB



After · gray-world white balance

Example 31421v: estimated RGB means (0.475, 0.481, 0.580); RGB gains (1.078, 1.065, 0.883).

Visual observation: the blue-purple cast on the building in this example is reduced. This is a qualitative preference, not a comparison against a calibrated reference.

Limitation: green vegetation, blue water or a dominant colored surface can violate the gray-world assumption. Neutralizing their average can remove a real scene color. Gain limits may prevent exact neutrality, and common highlight scaling may darken the image. Because the scans have no calibrated camera response or known reference colors, this is a display-space visual correction, not a physical recovery of the historical illuminant.

When gray-world is less convincing



Before · aligned RGB



After · gray-world white balance

In this river landscape, green vegetation and blue water contribute strongly to the average. The adjustment boosts red and suppresses green, leaving a pinker sky and a less natural-looking overall balance. This illustrates why equal channel means do not guarantee correct colors. The same fixed algorithm is retained for every scan; this failure is shown rather than tuned away.

Combined results

These pairs show the complete sequence on three examples. The baseline remains available for judging whether the result is preferable; improved contrast or a neutral mean alone does not establish historical color accuracy. Click an enhanced image for its original-pixel-scale result.

00163v



Before · aligned RGB



After · crop → white balance → contrast

00458u



Before · aligned RGB



After · crop → white balance → contrast

31421v



Before · aligned RGB



After · crop → white balance → contrast

Measurements across all 20 scans

Crop margins are measured in aligned-image pixels, ordered left / top / right / bottom. Gains are estimated after cropping for the combined pipeline and ordered R / G / B.

Plate	Detected margins	Area retained	Gray-world gains
00056v provided	23 / 1 / 16 / 11	86.9%	1.017 / 0.990 / 0.993
00125v provided	15 / 16 / 17 / 0	87.5%	1.223 / 0.858 / 0.984
00163v provided	9 / 10 / 13 / 8	89.3%	0.991 / 0.989 / 1.020
00458u provided	148 / 126 / 161 / 26	87.2%	0.999 / 1.005 / 0.996
00804v provided	21 / 8 / 16 / 5	87.0%	1.003 / 0.947 / 1.056
01007a provided	206 / 108 / 205 / 15	85.6%	0.889 / 1.020 / 1.117
01047u provided	33 / 87 / 161 / 267	84.1%	0.865 / 1.150 / 1.027
01164v provided	19 / 14 / 18 / 5	85.4%	1.107 / 0.932 / 0.977

Plate	Detected margins	Area retained	Gray-world gains
01269v provided	23 / 8 / 17 / 2	87.1%	0.769 / 0.968 / 1.502
01522v provided	17 / 8 / 20 / 5	86.9%	0.889 / 0.938 / 1.236
01597v provided	26 / 0 / 17 / 11	86.1%	0.937 / 0.981 / 1.094
01598v provided	23 / 0 / 18 / 1	89.2%	0.845 / 1.022 / 1.193
01657u provided	171 / 97 / 120 / 13	89.0%	0.883 / 1.147 / 1.004
01725u provided	63 / 0 / 119 / 0	95.0%	0.776 / 1.074 / 1.280
01728v provided	19 / 10 / 24 / 0	86.4%	0.884 / 1.007 / 1.142
01861a provided	82 / 0 / 157 / 66	91.5%	0.888 / 0.995 / 1.151
10131v provided	18 / 14 / 24 / 0	85.6%	0.869 / 0.975 / 1.214
31421v provided	11 / 0 / 20 / 5	90.8%	1.088 / 1.071 / 0.872
00166v additional	28 / 27 / 34 / 22	71.7%	0.950 / 0.909 / 1.180
00637v additional	20 / 5 / 13 / 12	86.8%	0.841 / 1.063 / 1.150

All 20 scans: baseline, crop, contrast, white balance and combined · Parameters and measured results (JSON)

07 / DISCUSSION & LIMITATIONS

Alignment is only part of restoration.

Visual assessment: the reconstructed previews show broadly aligned scene structure. This is a qualitative inspection, not a ground-truth accuracy score. The saved high-resolution outputs remain available for closer inspection.

Damaged exposures: the first detailed experiment has visible scratches and blotches in its individual grayscale channels. Their disagreement produces colored speckles after composition. Searching for a different global displacement cannot reconstruct these missing details.

Local residuals: faces, foliage and object boundaries can retain colored fringes. Motion between sequential exposures, channel-dependent brightness and departures from pure translation are plausible causes; the current experiment does not isolate their individual contributions. The model cannot correct local motion, rotation or scale change.

Border artifacts: excluding borders from the score prevents them dominating the alignment, but does not restore or remove the physical borders in the final image. The common-support crop only removes wrapped pixels.

Enhancement scope: the Bells & Whistles section implements detected-border cropping, contrast enhancement and gray-world white balance. The earlier L2/NCC galleries retain their original intensities for comparison. These post-processing methods cannot repair missing emulsion or local misalignment, and a more vivid result is not evidence of historically accurate color.

Dataset discrepancy: the handout mentions both 5 and 20 examples and legacy image names; the supplied directory contains 18 numbered JPEGs. All 18 were processed, and three selected plates receive detailed experiments.

08 / REPRODUCIBILITY

Measurements & reproducibility.

Seven synthetic-image unit tests cover channel order, 16-bit normalization, L2/NCC shift recovery, large displacements on odd dimensions, RGB composition, valid-support cropping, blank-image tie handling and invalid inputs. No mock tests are used.

Six additional unit tests exercise variable-width black, white and colored borders, images without borders, shared contrast endpoints, recovery of a known gray-world cast, finite outputs for constant and zero-channel images, and invalid inputs. The enhancement batch also checks source hashes and finite output ranges for every scan.

Run instructions are in the project README. Saved records include input SHA-256 hashes, channel dimensions, discarded rows, offsets, per-level scores and timings. NCC/L2 scores at different pyramid levels use different pixels and should not be interpreted as one comparable accuracy curve.

Environment: Python 3.13.5, NumPy 2.3.3, Pillow 11.1.0. Measurements were collected with one sequential batch, not a statistical performance benchmark.

Design & image sources.

1. **Website design reference:** Wix — AI Blog (Abstract). Visual reference for the independently implemented static website.
2. **Additional photograph:** V Malorossii (In Little Russia [i.e. Ukraine]). Prokudin-Gorskii photograph collection, Library of Congress, Prints and Photographs Division.
3. **Additional photograph:** Probnaia shtolnia (Trial adit; iron ore piled in front). Prokudin-Gorskii photograph collection, Library of Congress, Prints and Photographs Division.